

A PRIMER OF WATER STRUCTURING AND CATION ASSOCIATION IN CELLS

II. HISTORICAL NOTES, PRESENT STATUS, AND FUTURE DIRECTIONS

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Historically, many suggestions to the effect that the water in living cells has structure are to be found in the older literature, along with indirect supportive evidence.¹⁻⁵ As for the membrane theory of the cell, some authoritative historical notes on its development have been provided by Gilbert N. Ling.^{6,7}

Ling, however, is among the foremost protagonists of the new biophysics. Indeed, the modern history of the salt and water revolution in western science may be said to have begun with Ling, who invented the intracellular microelectrode while a graduate student at the University of Chicago in 1948.

SEEDS OF REVOLUTION

Ling's observation soon forced him to conclude that the generally accepted salt and water biophysics for the study of which he had invented the electrode was full of contradictions and therefore had to be wrong. He proceeded to construct a new and elegant model of the cell, incorporating the ideas that (A) cell water is structured, (B) cell cations are associated mostly with macromolecules, and (C) cation pumps do not exist.⁸ By 1949, Ling⁹ already had shown that cation pumps were thermodynamically impossible. Yet because he was in total disagreement with the vested scientific community, Ling had little success at the time in publishing either his conceptual or his experimental work in scientific journals. In the main, publication was delayed until 1960 when his collected studies appeared as a book.¹⁰

The new salt and water biophysics had an independent and virtually simultaneous beginning in Russia at the Institute of Cytology in Leningrad. There concepts analogous to those of Ling were initiated by Nasonov and summarized in a book which has been translated into English.¹¹

Following Nasonov, the Russian research was bellwethered by Troshin.¹² His investigations and those of Ling¹³⁻¹⁵ proceeded along independent but parallel paths. Both studied the equilibrium concentrations of various electrolytes and nonelectrolytes inside vs. outside of cells. Both showed that most solutes followed modified Langmuir binding isotherms, which implies one fraction of solute associated with macromolecules plus a second fraction dissolved in structured intracellular water.

By 1970, thanks to ingenious application of classical methods of analysis, the evidence for structured cell water associated cations in cells had become very strong.

CONTRIBUTIONS OF NMR

Meanwhile, beginning in 1965, nuclear magnetic resonance (NMR) provided much confirmation and some extension of the evidence being gathered by experimental observation that utilized the classical methods mentioned above. With regard to cell water structuring, Bratton et al.¹⁶ in 1965 first measured by pulsed NMR the shorter relaxation times of hydrogen of muscle water as compared with hydrogen of liquid water. Those measurements have since been confirmed repeatedly. With regard to cation association, Cope¹⁷⁻¹⁹ in 1965 first performed NMR analysis of ^{23}Na in tissues, demonstrating that within cells it behaves like Na^+ associated with charged sites on an ion exchanger resin rather than like Na^+ in free solution. Unlike the work of the Bratton group, the NMR measurements of Na^+ by Cope¹⁷⁻¹⁹ were intended specifically to test the concepts of Ling. Cope's approach stemmed and were initiated from certain solid-state physical concepts he had developed to account for electron transport in cells.^{20,21} He observed that cellular electron transport, in the light of those concepts, showed analogies with cellular cation transport. From this he reasoned that the analogies could be valid only if Ling's concepts of cation association were valid.^{22,23}

These ground-breaking studies triggered an explosion of fruitful NMR research. It began in 1969 as Hazlewood et al.²⁴ and Cope,²⁵ using different NMR techniques, independently and approximately at the same time showed an approximate 80-20% split of cell water into two fractions. Both fractions exhibited more structure than liquid water! The proliferation of significant NMR findings continued with demonstration by Cope and Damadian²⁶⁻²⁹ that ^{39}K in cells behaves not like K^+ in free solution but like K^+ associated with an ion exchanger resin—with the discovery by Damadian,^{30,31} that the NMR relaxation time of cell H_2O can be used for diagnosis of cancer—and with confirmation of those results by many other workers in the United States and abroad.

ADVANCES IN CONCEPT

The thermodynamic impossibility of cation pumps in the cell was given further evidence in muscle by Ling¹⁴ in 1969. Then in 1973 came striking demonstration of that impossibility by Minkoff and Damadian in bacteria.³²

Moreover, significant conceptual advances occurred as Damadian³³⁻³⁵ pointed out multiple analogies of the salt and water behavior of the cell to that of an ion exchanger resin granule. Investigation prompted by those parallels led to the dis-

covery by Minkoff and Damadian³⁶ of a probable contractile protein in bacteria as the underlying mechanism for K^+ accumulation. This in turn led to the concept that the experimentally observed preference of cell association sites for K^+ over Na^+ is controlled by restriction of concentration of intracellular water due to extrusion of water from the cell by shortening of the contractile fibers of the cell (a condition termed *cytotonus* by Damadian³⁶).

THE MODERN POSITION

The status of the new cellular salt and water biophysics at the present moment in its history would seem to be (as of September 1976):

1. The evidence is overwhelming for structured cell water, for associated Na^+ and K^+ in cells, and for the absence of cation pumps. The cell is to be regarded as an ion exchanger resin granule whose association sites on proteins have a marked preference for K^+ over Na^+ . During excitation, this preference is reduced by a configurational change of the proteins. But it is reestablished by ATP, which restores the resting state configuration of the proteins.

2. More work to prove the presence of structured water and associated cations in cells is not needed. That presence has been adequately proven. The selectivity of association sites for K^+ over Na^+ is well demonstrated.^{13-15,37,38} The important question now is what physical mechanism accomplishes that selectivity. To date, two plausible mechanisms have been proposed. The first is displacement of electrons within the cellular protein, as set forth in the association-induction hypothesis of Ling.^{10,14} The second proposed mechanism is cell water restriction by *cytotonus*, mentioned above as advocated by Damadian.³⁶ Solving the problem of mechanisms, whether by experimental or theoretical approaches, seems likely to remain difficult.

3. The old theoretical approaches of salt and water biophysics based on dilute solutions, pumps, and channels have been demonstrated to be not valid. They have been replaced by three new theoretical approaches: (a) the association-induction hypothesis of Ling,^{10,14} (b) the ion exchanger resin *cum* *cytotonus* theory of Damadian,^{32,36} and (c) the solid-state physical theory of Cope.²⁰⁻²³ All three approaches are based on the physicochemical realities of structural water and associated cations. They differ, however, in their respective approximations of reality and therefore in their methods of mathematical analysis.

The *association-induction hypothesis* relies on statistical-mechanical analysis. Such analysis can be exact and all-inclusive. But it is so complicated mathematically that equations in a form simple enough for experimental verification are difficult to obtain.

The *ion exchanger resin theory* is based on classic thermodynamics and electro-

static theory. Analysis here is by simpler mathematics. But it is probably not suitable for handling cooperative interactions among association sites.

The *solid-state theory* is quite simple mathematically. It has yielded simple predictions regarding cation leakage from cells, predictions that have been verified experimentally. But the theory is probably too simple to deal with the more complicated aspects of cation transport and excitability.

4. It follows from the above that the more complex applications of the new salt and water biophysics remain to be developed. Thus the phenomena now of greatest challenge to the new-breed research include excitability (with the objective of replacing Hodgkin-Huxley theory)—and cation transport across multicellular membranes; e.g., frog skin.³⁹

5. Despite admitted gaps in knowledge, the new salt and water biophysics has developed to the point where it stands ready for utilization in medical diagnosis and therapy. In the past, medical salt and water therapy has been entirely empirical. This was so because the old model of salt and water metabolism was not based on physico-chemical reality and therefore had no predictive value. To this author's knowledge, the still entrenched yet outworn and outdated concepts of cellular metabolism, involving "pumps" and the like, have never produced any useful medical results. The new biophysics (specifically the ion exchanger resin of Damadian³³⁻³⁵) has already led to a fast and accurate method for cancer diagnosis in tissue biopsies.³⁰⁻³¹ The method is being widely studied. NMR methods have been used by Damadian⁴⁰ to show a cross-sectional image of a cancer in an intact animal. At present, his group is experimentally attempting to apply the technique to cancer diagnosis (and perhaps treatment) in humans. Applications to other diseases await the interested investigator.

In sum, three fruitful approaches currently characterize research in the new and revolutionary biophysics. The three are the association-induction hypothesis, the ion exchanger resin theory, and the solid-state physical theory. These approaches, together with corollary avenues of investigation, are being pursued by a growing number of workers in the new biophysics. The harvest of data will undoubtedly fill out present theoretical constructs and engender new ones, at the same time contributing richly to scientific advance in general and clinical progress in particular.

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